Chapter 1

VIRTUAL WIND TUNNEL SETUP FOR CFD SIMULATIONS

In this section, model setup for the virtual wind tunnel with CFD is described. The following are the tasks conducted

1. Creation of CFD model Geometry.
2. Specifying solver informations for fluid and strcutral domain.
3. Finding the optimal mesh size for the Fluid domain with Mesh Converegnce Study

CFD Model:

The 2-D CFD model is created using GiD (Figure 1.3). Since forced motion oscillation tests are conducted, the structure, ie the bridge deck cross-section is modelled as a obpaque object in the fluid domain without any physics parameters like Stiffness and mass. Hence it is a one-way coupling. The bridge deck cross-section is taken from Sarkic (Reference1), shown in Figure 1.2. It is then externally oscillated by the solver with a specific frequency. Arbitrary Lagrangian-Eulerian (ALE) framework is used for this.

The Fluid domain is meshed with finer meshes near the structure and with coarser meshes farther away to have a optimal meshing. Hence, to produce a smoother mesh transitions, fluid domain is divided into five subregions as shown in Figure 1.3

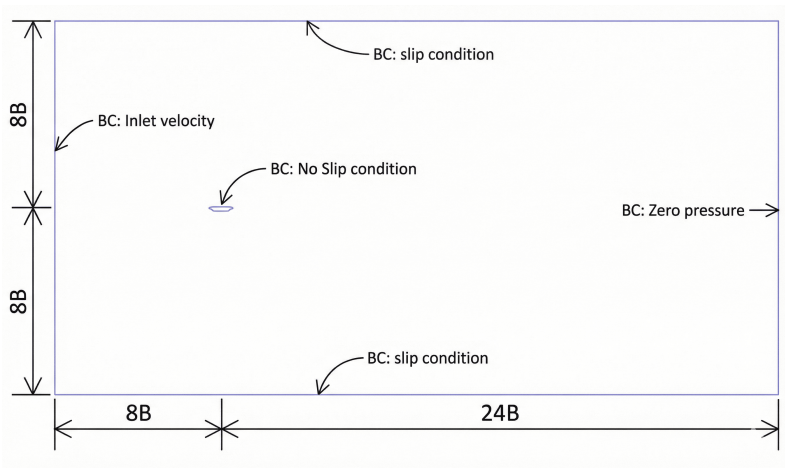


Figure 1.1: Virtual Wind Tunnel Dimensions

Solver Informations:

The GiD model is solved with an open source Multiphysics solver - KratosMultiphysics. The boundary conditions applied to Fluid domain are shown in Figure 1.1. The turbulence is modeled using an Unsteady Reynolds-Averaged Navier-Stokes (URANS) approach with the k–ω SST turbulence model, a turbulence intensity of 0.1, and a tur-

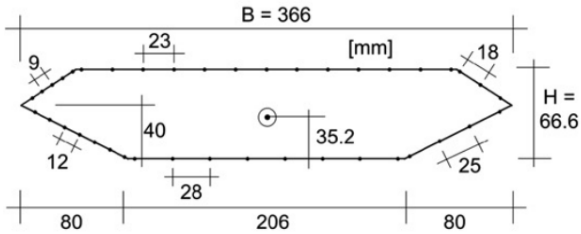


Figure 1.2: Sarkic Cross-Section

bulent mixing length of 0.0588 m.

The velocity is set to 1 m/s and the corresponding Reynolds number for the flow is $2.43 * 10^4$. The time step is taken as 0.002 seconds (Corresponding CFL is shown in Table 1.1). The total simulation time is divided into three phases: ramp-up time, stabilization time, and stable time period. In Ramp-up phase the flow velocity is steadily increased from 0 to 1 m/s to avoid numerical instabilities. In Stabilization phase, the flow is allowed to stabilize. The stable time period is the duration used for analysis. Ramp-up time and stabilization time are determined based on the vortex shedding frequency of the section. From EN 1991, the Strouhal number of the section is taken as 0.1.

Vortex shedding frequency $= \frac{0.1*1}{0.366} = 0.273$ Hz
Ramp up time $= 2 * \frac{1}{0.273} = 7.32$ seconds
Stabilization time $= 2 * \frac{1}{0.273} = 7.32$ seconds

Since the interest of this thesis is to capture the flutter response, stable time period is time taken for 10 complete forced motion oscillations. But as each reduced velocity has a different forced motion frequency, the stable phase of the simulations is different. A higher reduced velocity will have a lower oscillating frequency and hence requiring a higher time of simulation and vice versa. Hence its calculation is shown after the information about the prescribed motion frequency is provided in the consecutive chapters.

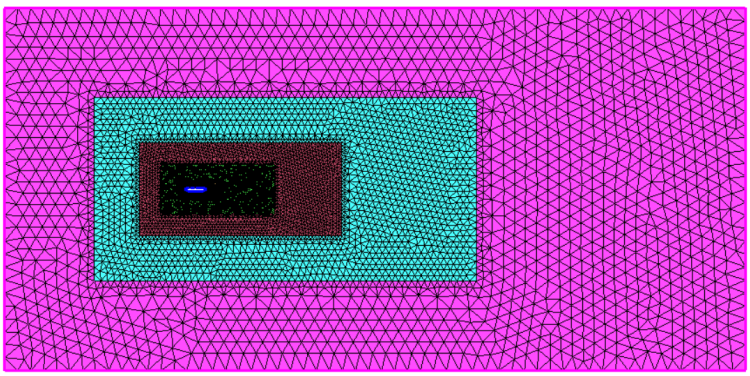


Figure 1.3: GiD Model of Virtual Wind Tunnel

Mesh Convergence Study:

A mesh convergence study is conducted to determine the optimal mesh element size. Five models with progressive mesh sizes are created and converegence of 8 derivatives are analysed. The Table 1.1 shows the models and corresponding mesh sizes at each zone. The study is performed for a single point on the flutter derivative curve at a reduced velocity of 10.93. From the literature, it is identified 5% of width of bridge deck as the heave amplitude and 5 degree as pitch amplitude are the appropriate forced motion amplitude such that aerodynamic forces show linear behaviour.

Heave amplitude = $\frac{5*0.366}{100}$ = 0.0183 m
Pitch amplitude = 0.087 radians
Motion Frequency = 0.25 Hz
Stable time period = $10 * \frac{1}{0.25}$ = 40 seconds
Total simulation time = Ramp-up time + Stabilization time + Stable time period = 7.32 + 7.32 + 40 = 54.64 seconds

Table 1.1: Mesh Convergence Study Results

	Model 1	**Model 2**	**Model 3**	**Model 4**	**Model 5**
Zone A (m)	0.25	0.25	0.2304	0.2112	0.192
Zone B (m)	0.1	0.12	0.1152	0.1056	0.096
Zone C (m)	0.054	0.08	0.0576	0.0528	0.048
Zone D (m)	0.018	0.025	0.0288	0.0264	0.024
Zone E (m)	-	-	0.0096	0.0088	0.008
Zone F (m)	0.009	0.008	0.0048	0.0044	0.004
Deck Surface (m)	0.0045	0.003	0.0024	0.0022	0.002
Width/ min mesh size	81.333	122	152.5	166.36	183
Total Elements	7368	15704	20854	24374	29408
Minimum Mesh Size-GID	0.00358	0.00245	0.00196	0.00179	0.00159
Velocity (m/s)	1	1	1	1	1
Time step	0.004	0.002	0.002	0.002	0.002
CFL	1.11	0.81	1.02	1.11	1.26

With this simulation setup, the CFD simulations are conducted for all five models Simulations Result of the Model 3 is shown in figure 1.4. From this result, pressures on the structural surface is integrated to obtain aerodynamic lift and moment forces. Figure 1.5 shows the aerodynamic lift and moment forces obtained for Model 3. These aerodynamic forces are then used to calculate the flutter derivatives. The convergence of flutter derivatives is then analysed.

The results of the mesh convergence study are shown in Figure 1.6. It is observed that the derivatives H_1, H_3, A_1, A_2, and A_3 exhibited good convergence, while derivatives H_2, H_4, and A_4 displayed oscillatory convergence behavior. This behavior is consistent with findings in the literature, as these particular derivatives are known to be more difficult to capture accurately. From the study it is concluded, Model 3 is optimal and this mesh

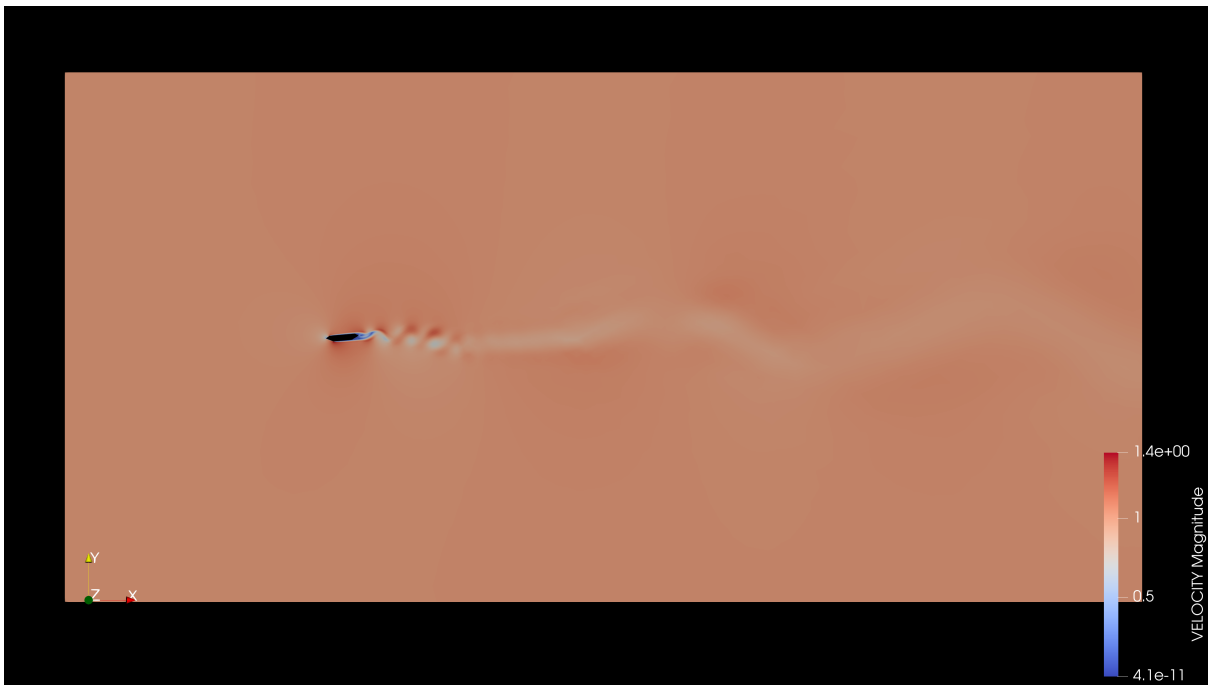
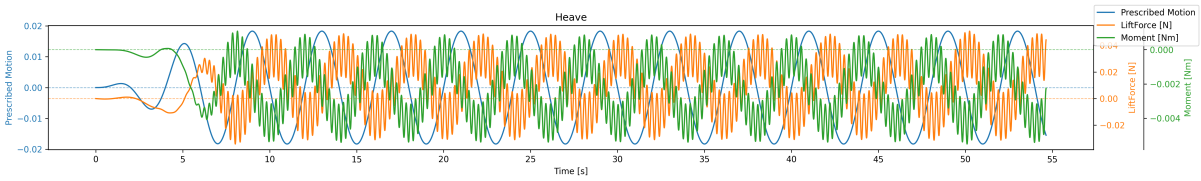
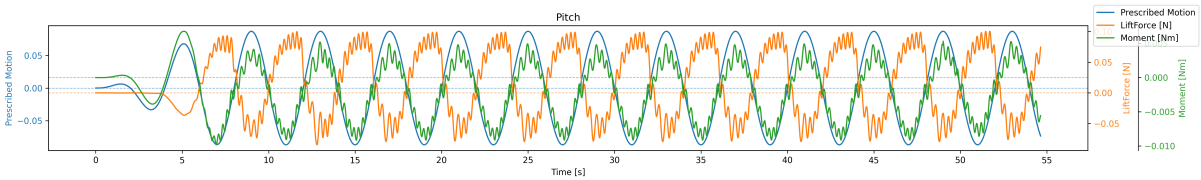


Figure 1.4: CFD Results for Model 3



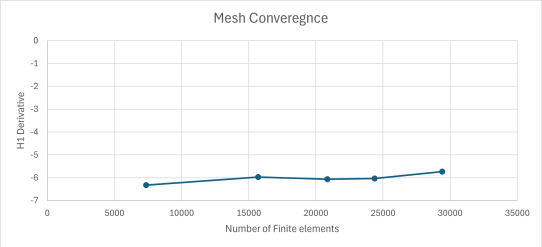
(a) Heave



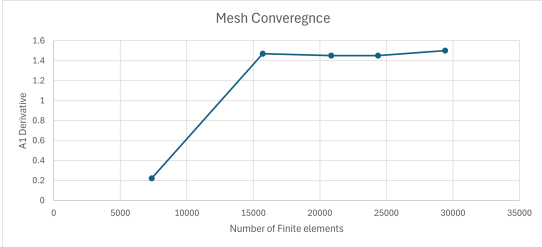
(b)
Pitch

Figure 1.5: Heave Motion

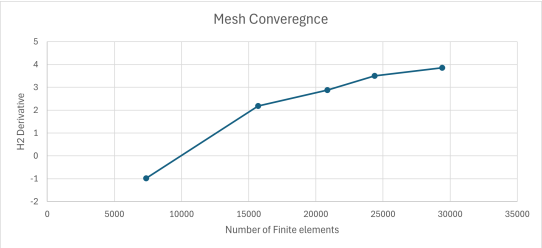
size is used for all subsequent simulations.



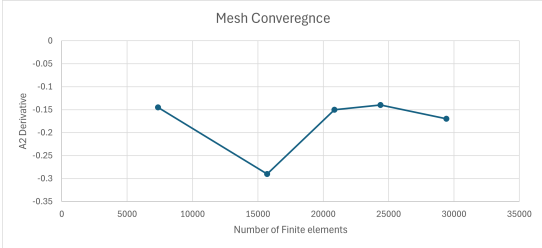
(a) H1



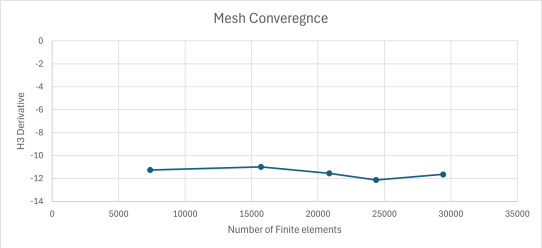
(b) A1



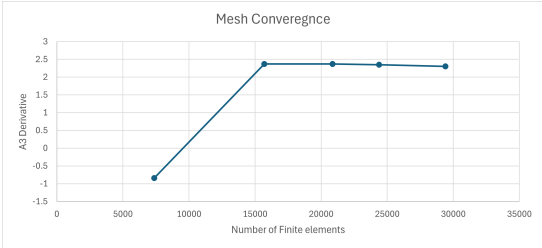
(c) H2



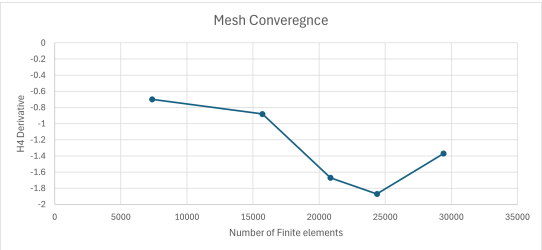
(d) A2



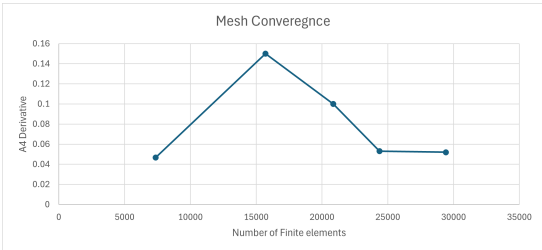
(e) H3



(f) A3



(g) H4



(h) A4

Figure 1.6: Mesh Convergence Study Results